Climate Informatics 2026 (2026), xx: 1–2
doi:10.1017/xxxx

CAMBRIDGE UNIVERSITY PRESS

CLIMATE INFORMATICS 2026
EXTENDED ABSTRACT

Evaluating Spatial Generalisation of Generative Downscaling

Maybritt Schillinger[1]*, Luca Schmidt[2] and Nicolai Meinshausen[1]

[1]Seminar for Statistics, ETH Zurich, Zurich, Switzerland
[2]Cluster of Excellence Machine Learning, University of Tuebingen, Tuebingen, Germany
*Corresponding author. Email: maybritt.schillinger@stat.math.ethz.ch

Received xx xxx xxxx

Keywords: climate informatics, downscaling, domain generalisation

1. Introduction and objectives

Global high-resolution (HR) climate data is important for impact assessments and planning of adaptation strategies. Deep-learning based downscaling can be a key tool in providing such data. However, training these methods with observational data requires HR training data (e.g. radar/observational data), which is unevenly distributed around the globe. Similarly, downscaling trained with data from physics-based regional climate models (RCMs) is challenging because the availability of RCM datasets varies drastically across domains, with limited coverage for example in Asia or Africa [1].

These issues motivate machine-learning based downscaling approaches that are able to generalise across domains, using only low-resolution (LR) predictors and geographic features as inputs and [6, 3, 2].

2. Data and methods

We use global daily temperature and precipitation data from ERA5 as HR targets. LR predictors are obtained by average-pooling of the HR targets, which avoids potential dataset inconsistencies between LR inputs and HR targets that can arise if inputs and outputs stem from different datasets (e.g., [3]). In addition, we include orography and the land-sea mask from ERA5 as predictors.

We build on *EnScale*, a multivariate generative model developed to emulate RCMs from global climate model (GCM) data [5]. It consists of a coarse-correction step, which maps GCM inputs to average-pooled RCM data, followed by a *super-resolution* step, which maps average-pooled RCM data to RCM targets. The *super-resolution* map performs multiple resolution enhancement stages, using layers that combine weights shared over space with weights specific to each location, and are therefore not fully convolutional.

We adapt the *EnScale* model for our setting as follows. Firstly, we use only the *super-resolution* component. Secondly, we incorporate the geographic predictors into the model layers. For this, orography from each location and its local neighbourhood is mapped into a learnable latent representation, which is then concatenated with the LR predictors and passed to the *EnScale* layers. Finally, we replace the location-specific weights with weights that are shared across space.

© The Authors(s) 2020. Published by Cambridge University Press. This is an Open Access article, distributed under the terms of the Creative Commons Attribution licence (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted re-use, distribution, and reproduction in any medium, provided the original work is properly cited.

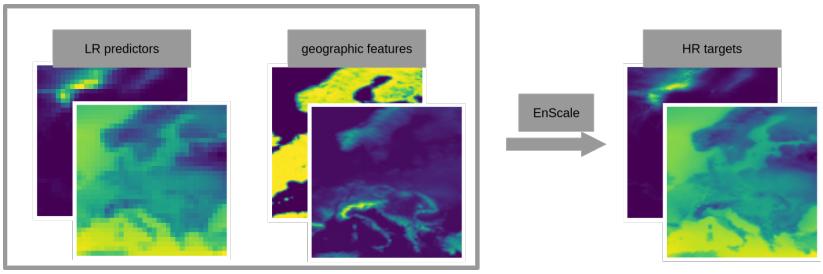

Figure 1. *Inputs (LR predictors and geographic features) and HR targets for downscaling with EnScale. The figure shows temperature and precipitation data for one exemplary patch in Europe on the first of January 1950.*

3. Expected outcomes

We aim to answer the following questions:

1. Can a fully convolutional architecture based only on LR inputs and orography match the performance of the original *EnScale* architecture, which uses location-specific layers?
2. Can this model generalise to regions not seen during training, and which factors determine generalisation performance?
3. At which target resolutions do geographic features provide the most added value, and how does spatial generalisation performance change with higher/lower target resolution?

4. Discussion

Our above research questions overlap with those addressed in previous work. Existing literature, however, uses ML models such as GANs and diffusion models [4, 2, 3], whereas we focus on the simpler and computationally more efficient *EnScale* model. Since *EnScale* splits the downscaling task into multiple simpler resolution-refinement steps, it naturally allows for investigations of various target resolutions.

For spatial generalisation, previous work highlights example cases where generalisation works well or poorly. We aim to investigate this more systematically, e.g., by better characterising differences between regions.

References

[1] *CORDEX Website*. URL: https://cordex.org/data-access/cordex-cmip5-data/cordex-cmip5-simulations/.

[2] Paula Harder et al. "On Global Applicability and Location Transferability of Generative Deep Learning Models for Precipitation Downscaling". In: (Dec. 2025). URL: http://arxiv.org/abs/2512.01400.

[3] Paula Harder et al. "RainShift: A Benchmark for Precipitation Downscaling Across Geographies". In: (July 2025). URL: http://arxiv.org/abs/2507.04930.

[4] Ayush Prasad et al. "Evaluating the transferability potential of deep learning models for climate downscaling". In: (July 2024). URL: http://arxiv.org/abs/2407.12517.

[5] Maybritt Schillinger et al. "EnScale: Temporally-consistent multivariate generative downscaling via proper scoring rules". In: (Dec. 2025). URL: http://arxiv.org/abs/2509.26258.

[6] Yingkai Sha et al. "Deep-learning-based gridded downscaling of surface meteorological variables in complex terrain. Part i: Daily maximum and minimum 2-m temperature". In: *Journal of Applied Meteorology and Climatology* 59.12 (Dec. 2020), pp. 2057–2073. ISSN: 15588432. DOI: 10.1175/JAMC-D-20-0057.1.